

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277355

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Kim; Dongsoo et al.

CONSTRUCTION EQUIPMENT

Abstract

A construction equipment includes a lower traveling body, an upper swing body on the lower traveling body, a work device comprising a boom, an arm, and a bucket moved by respective hydraulic cylinders and supported by the upper swing body, a control valve for controlling the hydraulic cylinder, an operation lever for outputting an operation signal, a work setting unit for setting and/or selecting the work area of the work device, a location information provision unit, and an electronic control unit (ECU). The ECU calculates the distance between the work area and the work device and controls the speed of the work device on the basis of the calculated distance. The ECU sets the speed limit range and/or speed reducing rate on the basis of the shortest distance between the bucket end and the work area in a reference bucket pin location.

Inventors: Kim; Dongsoo (Gyeongsangnam-do, KR), Kim; Miok (Gyeongsangnam-do, KR)

Applicant: Volvo Construction Equipment AB (Eskilstuna, SE)

Family ID: 72666539

Appl. No.: 19/211507

Filed: May 19, 2025

Related U.S. Application Data

parent US continuation 17601167 20211004 PENDING WO continuation PCT/KR2019/004110
20190405 child US 19211507

Publication Classification

Int. Cl.: E02F9/26 (20060101); E02F3/43 (20060101); E02F9/22 (20060101)

U.S. Cl.:

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/601,167 filed Oct. 4, 2021, which is a 35 U.S.C. § 371 national stage application of PCT International Application No. PCT/KR2019/004110 filed on Apr. 5, 2019, the entire contents of which are incorporated by reference herein in their entireties.

TECHNICAL FIELD

[0002] The present invention relates to a construction equipment. More specifically, the present invention relates to a construction equipment having a work area limit function capable of improving work speed and work efficiency by controlling the speed of the bucket in consideration of the distance from the bucket end to the work area in the direction in which the bucket end moves (the speed direction of the bucket end).

BACKGROUND

[0003] In general, an excavator is a construction equipment performing various tasks such as digging for digging up the ground at construction sites, etc., loading for carrying soil, excavating for making a foundation, crushing for dismantling buildings, grading for cleaning the ground, and leveling for leveling the ground.

[0004] With reference to FIG. 1, a construction equipment 1 such as an excavator comprises a lower traveling body 2, an upper swing body 3 rotatably supported on the lower traveling body 2, and a work device 4 installed to operate vertically on the upper swing body 3.

[0005] In addition, the work device 4 is formed of multi-joints, and comprises a boom 4a which has a rear end thereof rotatably supported on the upper swing body 3, an arm 4b which has a rear end thereof rotatably supported on a tip of the boom 4a, and a bucket 4c rotatably installed on the tip side of the arm 4b. Also, hydraulic oil is provided as the user operates the lever, and a boom cylinder 5 (working actuator), an arm cylinder 6 (working actuator), and a bucket cylinder 7 (working actuator) operate the boom 4a, arm 4b, and bucket 4c, respectively.

[0006] Such construction equipment 1 operates a work device 4 such as a boom 4a, an arm 4b, a bucket 4c, etc. by its respective manual operation lever. However, since the work device 4 carries out a rotational movement by being connected to each joint part, it requires considerable effort for the driver to operate each work device 4 separately and work in a predetermined area.

[0007] Therefore, in order to facilitate such work, Japanese Patent No. Hei7-94735 discloses an apparatus for controlling work area of an excavator. The apparatus for controlling work area of an excavator controls the movement of the bucket along the distance between the bucket end and the border line of a non-intrusive area. Therefore, even if the driver accidentally moves the bucket end to the non-intrusive area, the bucket will automatically stop at the border line of the non-intrusive area. Also, the driver may realize that the work device is approaching the non-intrusive area from the fact that the speed of the work device is decreasing during work, and send back the tip of the bucket.

[0008] FIG. 2 illustrates situations (a) to (c) in which the posture of the bucket 13c is different, but the distance between the bucket end P1 and the work area is the same in a situation in which a driver sets a work area and then removes the work material piled up in the work area.

[0009] In this case, it is necessary to operate a bucket 13c in order for a driver to remove the work material piled up in the work area set. According to prior art, the speed of the bucket 13c is limited only by the shortest distance d between the bucket end P1 and the work area. Even when the shortest distance d between the bucket end P1 and the work area is the same, there may be many cases where the bucket 13c does not invade the work area or has extra time to invade the work area

depending on the posture of the bucket **13c**. Even in such cases, according to prior art, the speed of the bucket **13c** is limited all the same.

[0010] In other words, unlike situations (a) and (b), even though situation (c) is a situation in which the bucket **13c** does not invade the work area when the driver operates the bucket **13c**, the shortest distance d between the bucket end **P1** and the work area is recognized to be the same, and thus the speed of the bucket **13c** is limited in the same manner as in situations (a) and (b). Accordingly, there is a problem that the work speed and efficiency may decrease during the digging work using a bucket **13c**.

SUMMARY

[0011] The present invention aims at solving the above problems of the prior art. It is an object of the present invention to provide a construction equipment having a work area limit function capable of improving work speed and work efficiency by controlling the speed of the bucket in consideration of the distance from the bucket end to the work area in the direction in which the bucket end moves (the speed direction of the bucket end).

[0012] In order to achieve the above object, according to an aspect of the present invention, the present invention provides a construction equipment comprising: a lower traveling body; an upper swing body rotatably supported on the lower traveling body; a work device comprising a boom, an arm, and a bucket moved by means of respective hydraulic cylinders and supported by the upper swing body; a control valve for controlling the hydraulic cylinder; an operation lever for outputting an operation signal corresponding to the operation amount of a driver; a work setting unit capable of setting and/or selecting the work area of the work device; a location information provision unit for collecting and/or calculating location information of the work device and/or location information of the work area; and an electronic control unit for outputting a control signal for the control valve according to a signal inputted from at least one from among the control lever, the work setting unit, and the location information provision unit, wherein the electronic control unit is configured to calculate the distance between the work area and the work device and to control the speed of the work device on the basis of the calculated distance.

[0013] According to an embodiment of the present invention, the electronic control unit may determine whether the work device approaches or moves away from the work area when an operation signal of the operation lever is input, and limit the speed of the work device only when the work device approaches the work area.

[0014] According to an embodiment of the present invention, the electronic control unit may control the speed of the work device on the basis of the distance from the bucket end to the work area in the speed direction of the bucket end.

[0015] According to an embodiment of the present invention, the direction in which the bucket end moves may be a direction in which the line connecting the bucket end and the bucket pin is perpendicular to the bucket end.

[0016] According to an embodiment of the present invention, the direction in which the bucket end moves may be a direction in which the line connecting the bucket end and the arm pin is perpendicular to the bucket end.

[0017] According to an embodiment of the present invention, the electronic control unit may compare the distance from the bucket end to the work area in the speed direction of the bucket end with the predetermined reference value and determine to be in the speed limit range when the distance from the bucket end to the work area in the speed direction of the bucket end is smaller than the predetermined reference value.

[0018] According to an embodiment of the present invention, the electronic control unit may set the speed reducing rate of the work device in the speed limit range and limit the speed of the work device on the basis of the set speed reducing rate.

[0019] According to an embodiment of the present invention, the electronic control unit may set the speed limit range and/or speed reducing rate on the basis of the shortest distance between the

bucket end and the work area in a reference bucket pin location.

[0020] According to an embodiment of the present invention, the electronic control unit may control the speed of the work device on the basis of the speed reducing rate when the work device enters the speed limit range.

[0021] According to an embodiment of the present invention, the electronic control unit may reduce the speed limit amount of the work device in proportion to the raised distance of the bucket pin when the location of the bucket pin is raised to be higher than the location of the reference bucket pin.

[0022] According to an embodiment of the present invention, the location information provision unit may comprise at least one of a location measuring unit for measuring the location information of the construction equipment, a posture measuring unit for measuring the posture information of the construction equipment and the location of the respective work device, and a coordinate calculating unit for calculating the coordinate on the basis of the location information measured from the location measuring unit and the posture measuring unit.

[0023] According to an embodiment of the present invention, the operation lever may generate an electric signal in proportion to the operation amount of a driver and provide the same to the electronic control unit as an electric joystick.

[0024] According to an embodiment of the present invention, the work setting unit may provide a plurality of work mode setting functions that can be set according to the driver's need, and display, on a display screen, at least one of the geographic information, location information and posture information of the construction equipment provided from the location information provision unit according to the work mode setting.

[0025] According to an aspect of the present invention, the work efficiency may be improved by controlling the speed of the work device on the basis of the distance from the bucket end to work area in the direction in which the bucket end moves.

[0026] In addition, the driver may easily operate the work device regardless of driving experience.

[0027] The effects of the present invention are not limited to the above-mentioned effects, and it should be understood that the effects of the present disclosure include all effects that could be inferred from the configuration of the invention described in the detailed description of the invention or the appended claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0028] FIG. 1 is a perspective view illustrating a basic configuration of the construction equipment according to prior art;

[0029] FIG. 2 is a schematic diagram illustrating a method for controlling the speed of the work device according to prior art;

[0030] FIG. 3 is a schematic diagram illustrating the work area limit function of the construction equipment according to an embodiment of the present invention;

[0031] FIG. 4 is a schematic diagram illustrating a method for controlling the speed of the work device during bucket-in operation according to an embodiment of the present invention;

[0032] FIG. 5 is a schematic diagram illustrating a method for controlling the speed of the work device during arm-in operation of the construction equipment according to another embodiment of the present invention; and

[0033] FIG. 6 is a schematic diagram illustrating a method for controlling the speed of the work device during bucket-in operation of the construction equipment according to yet another embodiment of the present invention.

DETAILED DESCRIPTION OF EMBODIMENTS

[0034] Hereinafter, embodiments of the present invention will be explained in detail with reference to FIGS. 1 to 6.

[0035] The construction equipment **10** according to an embodiment of the present invention comprises a lower traveling body **11**, an upper swing body **12** rotatably supported on the lower traveling body **11**, and a work device **13** supported by the upper swing body **12**. The work device **13** comprises a boom **13a**, an arm **13b**, and a bucket **13c** which operate by means of respective hydraulic cylinders.

[0036] Also, the construction equipment **10** according to an embodiment of the present invention has a work area limit function which controls the work device **13** not to invade the work surface by limiting the operation required amount of the driver's work device **13** on the basis of the distance between the bucket end and the work surface.

[0037] FIG. 3 is a schematic diagram illustrating the work area limit function of the construction equipment according to an embodiment of the present invention, and FIG. 4 is a schematic diagram illustrating a method for controlling the speed of the work device during bucket-in operation according to an embodiment of the present invention.

[0038] With reference to FIGS. 3 and 4, the construction equipment **10** having a work area limit function according to an embodiment of the present invention comprises a lower traveling body **11**, an upper swing body **12** rotatably supported on the lower traveling body **11**, a work device **13** comprising a boom **13a**, an arm **13b**, and a bucket **13c** moved by means of respective hydraulic cylinders and supported by the upper swing body **12**, a control valve **100** for controlling the hydraulic cylinder, an operation lever **200** for outputting an operation signal corresponding to the operation amount of a driver, a work setting unit **400** capable of setting and/or selecting the work area of the work device, a location information provision unit **300** for collecting and/or calculating location information of the work device and/or location information of the work area, and an electronic control unit **500** for outputting a control signal for the control valve **100** according to a signal inputted from at least one from among the operation lever **200**, the work setting unit **400**, and the location information provision unit **300**.

[0039] At this time, the electronic control unit **500** according to an embodiment of the present invention is configured to calculate the distance between the work area and the work device and to control the speed of the work device on the basis of the calculated distance.

[0040] The control valve **100** is a member for opening and closing the flow path by a spool moving in the axial direction under pressure. In other words, the control valve **100** serves to switch the supply direction of the hydraulic oil supplied by the hydraulic pump which is the hydraulic pressure source to the hydraulic cylinder side. The control valve **100** is connected to the hydraulic pump through a hydraulic pipe, and induces the supply of hydraulic oil from the hydraulic pump to the hydraulic cylinder.

[0041] The operation lever **200** may be a hydraulic joystick or an electric joystick, and preferably may be an electric joystick which generates an electric signal in proportion to the operation amount of a driver and provides the same to the electronic control unit **500**.

[0042] It is determined whether the work device **13** approaches or moves away from the set work area by the current location of the work device **13** and the operating direction of the operation lever **200**.

[0043] The speed of the work device **13** may not be limited when the work device **13** moves away from the work area.

[0044] The location information provision unit **300** may comprise at least one of a location measuring unit for measuring the location information of a construction equipment **10** by receiving a signal transmitted by a global positioning system (GPS) satellite, a posture measuring unit for measuring the posture information of the construction equipment **10** and the location of at least one of the boom **13a**, the arm **13b** and the bucket **13c**, and a coordinate calculating unit for calculating the coordinates of the construction equipment **10** on the basis of the location information measured

from the location measuring unit and the posture measuring unit.

[0045] The location measuring unit **310** may comprise a receiver capable of receiving a signal transmitted by a GPS satellite, and measure location information of the construction equipment **10** from the received signal.

[0046] The posture measuring unit **320** measures the location and/or posture of at least one of the boom **13a**, arm **13b** and bucket **13c** and slope of the body of the construction equipment **100** using a plurality of inertial measurement units (IMUs) and angle sensors, etc.

[0047] The coordinate calculating unit **330** calculates the coordinates (x, y, z) of at least one of the boom **13a**, arm **13b**, and bucket **13c** using the location information measured from the location measuring unit **310** and the posture measuring unit **320**.

[0048] In addition, the location information providing unit **300** may further comprise a mapping unit for mapping the geographic information around the work location and the construction information on the work location to the calculated coordinates. The mapping unit maps by adjusting the location and/or posture of the respective work device **13** measured by the posture measuring unit and the slope of the body of the construction equipment **10** according to each axis calculated by the coordinate calculating unit.

[0049] The work setting unit **400** may set and/or select a work area of the work device **13**. In addition, the work setting unit **400** may provide a work mode function that may be set and/or selected in various ways according to the driver's needs, such as work area limit mode, swing position control mode.

[0050] The work setting unit **400** displays at least one of the geographic information, location information and posture information of the construction equipment **10** provided from the location information provision unit **300** on a display **410** screen according to the setting and/or selection of the work area and/or the work mode. In other words, the driver may set and/or select a work area and/or a work mode on the screen of the display **410**, and work easily using the displayed information accordingly. In this case, the work area refers to a design surface targeted by the driver.

[0051] The electronic control unit **500** determines whether the current work device **13** approaches or moves away from a set work area when an operation signal is input from the operation lever **200**. When it is determined that the work device **13** is approaching a set work area, the distance between the work device **13** and the set work area is calculated. Then, the calculated distance is compared with a predetermined reference value to determine the speed limit of the work device **13**. Finally, a control signal is output to the control valve **100** for controlling the hydraulic cylinder on the basis of the speed limit.

[0052] In other words, when the work area limit function according to the present invention is active, the operation signal of the operation lever **200** and/or various location information of the location information provision unit **300** are input to the electronic control unit **500**. In addition, the electronic control unit **500** determines the speed limit amount of the work device **13** on the basis of the collected information, and controls the movement of the work device **13** accordingly.

[0053] With reference to FIGS. **3** and **4**, the construction equipment having a work area limit function according to an embodiment of the present invention operates as shown below.

[0054] First, the driver selects an active control mode on the work setting unit **400**, and sets a target work area. In addition, the driver operates the bucket-in operation lever **200** for bucket digging work on the work area.

[0055] At this time, whether the work device **13** approaches or moves away from a set work area is determined by the current location of the work device **13** and the operating direction of the operation lever **200**. The speed of the work device **13** may be limited only when the work device **13** approaches the work area.

[0056] Then, the location information provision unit **300** collects and/or calculates the location information of the work device **13** and/or the location information of the set work area, and provides the same to the electronic control unit **500**.

[0057] The electronic control unit **500** calculates the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the direction in which the bucket end **P1** moves (the speed direction of the bucket end) on the basis of the location information of the work device **13** and/or location information of the set work area provided from the location information provision unit **300**.

[0058] Here, the direction in which the bucket end **P1** moves may be a direction in which a virtual line (line **1**) connecting the bucket end **P1** and the bucket pin is perpendicular to the bucket end **P1**. In other words, it may be a direction facing a tangential line at the bucket end **P1** from a circle having a virtual line connecting the bucket end **P1** and the bucket pin **P2** as a diameter.

[0059] Next, the electronic control unit **500** compares the calculated distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1** with a predetermined reference value $d_{sub.t}$.

[0060] The electronic control unit **500** determines that the speed of the bucket **13c** does not have to be limited when the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1** is greater than the predetermined reference value $d_{sub.t}$. In other words, the electronic control unit **500** does not limit the speed of the bucket **13c** at this time.

[0061] However, when the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1** is smaller than the reference value $d_{sub.t}$, the electronic control unit **500** determines it to be in a speed limit range.

[0062] Then, the electronic control unit **500** sets the speed reducing rate of the bucket **13c** in the speed limit range. At this time, the speed reducing rate of the bucket **13c** may be linearly set according to the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1**, but is not limited thereto.

[0063] Accordingly, the speed of the bucket **13c** is controlled on the basis of the speed reducing rate in the speed limit range.

[0064] In other words, the electronic control unit **500** outputs a control signal to the control valve **100** on the basis of the speed reducing rate according to the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1**, and the control valve **100** controls the hydraulic cylinder on the basis of the control signal.

[0065] With reference to FIG. 4, a method for controlling according to the posture of the bucket during bucket digging will be described as shown below.

[0066] The shortest distance d between the bucket end **P1** and the work area is all the same in postures (a), (b) and (c), but the time it takes for each posture to invade the work area while operating the bucket **13c** is different.

[0067] Nevertheless, if the speed of the bucket **13c** is limited on the basis of the shortest distance d between the bucket end **P1** and the work area, the speed of the bucket **13c** is limited all the same in postures (a) to (c). Accordingly, the work speed and efficiency may decrease during the digging work using the bucket **13c**.

[0068] Therefore, the speed of the bucket **13c** needs to be controlled on the basis of the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1**.

[0069] At this time, in posture (a), the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1** is $d_{sub.1}$, which is the smallest, in posture (b), the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1** is $d_{sub.2}$, which is the second smallest, and in posture (c), the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1** is $d_{sub.3}$, which is the largest.

[0070] In other words, according to the present invention, in posture (a), the speed of the bucket **13c** is reduced the most, in posture (b), the speed of the bucket **13c** is reduced the second most, and in posture (c), the speed of the bucket **13c** is reduced the least.

[0071] As such, when the electronic control unit **500** controls the speed of the bucket **13c** on the basis of the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1**, it becomes possible to operate the bucket **13c** more efficiently than when the bucket

13c does not invade the work area or when there is a lot of time to invade the work area according to the posture of the bucket **13c** during bucket digging.

[0072] FIG. 5 is a schematic diagram illustrating a method for controlling the speed of the work device during arm-in operation of the construction equipment according to another embodiment of the present invention.

[0073] With reference to FIGS. 3 and 5, the method for operating the construction equipment having a work area limit function according to another embodiment of the present invention is as shown below.

[0074] First, a driver selects an active control mode on the work setting unit **400** and sets a target work area. In addition, the driver operates the arm-in operation lever **200** for digging work on the work area. At this time, the operation lever **200** of the bucket **13c** may not be operated.

[0075] At this time, whether the work device **13** approaches or moves away from a set work area is determined by the current location of the work device **13** and the operating direction of the operation lever **200**. The speed of the work device **13** may be limited only when the work device **13** approaches the work area.

[0076] Then, the location information provision unit **300** collects and/or calculates the location information of the work device **13** and/or the location information of the set work area, and provides the same to the electronic control unit **500**.

[0077] The electronic control unit **500** calculates the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1** on the basis of the location information of the work area **13** and/or the location information of the set work area provided from the location information provision unit **300**.

[0078] Here, the direction in which the bucket end **P1** moves may be a direction in which a virtual line (line **2**) connecting the bucket end **P1** and the arm pin **P3** is perpendicular to the bucket end **P1**. In other words, it may be a direction facing a tangential line at the bucket end **P1** from a circle having a virtual line connecting the bucket end **P1** and the arm pin **P3** as a diameter.

[0079] Next, the electronic control unit **500** compares the calculated distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1** with a predetermined reference value $d_{sub.t}$.

[0080] The electronic control unit **500** determines that the speed of the bucket **13c** does not have to be limited when the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1** is greater than the predetermined reference value $d_{sub.t}$. In other words, the electronic control unit **500** does not limit the speed of the bucket **13c** at this time.

[0081] However, when the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1** is smaller than the reference value $d_{sub.t}$, the electronic control unit **500** determines it to be in a speed limit range.

[0082] Then, the electronic control unit **500** sets the speed reducing rate of the arm **13b** in the speed limit range. At this time, the speed reducing rate of the arm **13b** may be linearly set according to the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1**, but is not limited thereto.

[0083] Accordingly, the speed of the arm **13b** is controlled on the basis of the speed reducing rate in the speed limit range.

[0084] In other words, the electronic control unit **500** outputs a control signal to the control valve **100** on the basis of the speed reducing rate according to the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1**, and the control valve **100** controls the hydraulic cylinder on the basis of the control signal.

[0085] With reference to FIG. 5, a method for controlling according to the posture of the arm **13b** during digging will be described as shown below.

[0086] The shortest distance d between the bucket end **P1** and the work area is all the same in postures (a), (b) and (c), but the time it takes for each posture to invade the work area while

operating the arm **13b** is different.

[0087] Nevertheless, if the speed of the arm **13b** is limited on the basis of the shortest distance d between the bucket end **P1** and the work area, the speed of the arm **13b** is limited all the same in postures (a) to (c). Accordingly, the work speed and efficiency may decrease during the digging work using the arm **13b**.

[0088] Therefore, the speed of the arm **13b** needs to be controlled on the basis of the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1**.

[0089] At this time, in posture (a), the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1** is $d_{sub.1}$, which is the smallest, in posture (b), the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1** is $d_{sub.2}$, which is the second smallest, and in posture (c), the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1** is $d_{sub.3}$, which is the largest.

[0090] In other words, in posture (a), the speed of the arm **13b** is reduced the most, in posture (b), the speed of the arm **13b** is reduced the second most, and in posture (c), the speed of the arm **13b** is reduced the least.

[0091] As such, when the electronic control unit **500** controls the speed of the arm **13b** on the basis of the distance $d_{sub.s}$ from the bucket end **P1** to the work area in the speed direction of the bucket end **P1**, it becomes possible to operate the arm **13b** more efficiently than when the bucket **13c** does not invade the work area or when there is a lot of time to invade the work area according to the posture of the arm **13b** during bucket digging.

[0092] FIG. **6** is a schematic diagram illustrating a method for controlling the speed of the work device during bucket-in operation of the construction equipment according to yet another embodiment of the present invention.

[0093] With regard to FIGS. **3** and **6**, the method for operating the construction equipment having a work area limit function according to yet another embodiment of the present invention is as shown below.

[0094] First, a driver selects an active control mode on the work setting unit **400** and sets a target work area. In addition, the driver operates the bucket-in operation lever **200** for digging work on the work area.

[0095] At this time, whether the work device **13** approaches or moves away from a set work area is determined by the current location of the work device **13** and the operating direction of the operation lever **200**. The speed of the work device **13** may be limited only when the work device **13** approaches the work area.

[0096] Then, the location information provision unit **300** collects and/or calculates the location information of the work device **13** and/or the location information of the set work area, and provides the same to the electronic control unit **500**.

[0097] The electronic control unit **500** sets the location of the reference bucket pin **P2** on the basis of the location information of the work device **13** and/or location information of the predetermined work area provided from the location information provision unit **300**. In addition, the speed limit range and/or the speed reducing rate are set on the basis of the shortest distance d between the bucket end **P1** and the work area at the location of the reference bucket pin **P2**.

[0098] The electronic control unit **500** controls the speed of the bucket **13c** according to a set speed reducing rate on the basis of the shortest distance d between the bucket end **P1** and the work area when the bucket end **P1** enters the speed limit range.

[0099] Here, when the location of the bucket pin **P2** moves from the location of the reference bucket pin **P2**, the electronic control unit **500** determines whether the location of the bucket pin **P2** is raised or lowered than the location of the reference bucket pin **P2**.

[0100] When it is determined that the location of the bucket pin **P2** is raised to be higher than the location of the reference bucket pin **P2**, the electronic control unit **500** may reduce the speed limit of the bucket **13c** in proportion to the distance the bucket pin **P2** is raised.

[0101] When it is determined that the location of the bucket pin P2 is lowered to be lower than the location of the reference bucket pin P2, the electronic control unit 500 may increase the speed limit of the bucket 13c in proportion to the distance the bucket pin P2 is lowered.

[0102] In other words, the electronic control unit 500 may set the speed limit range and the speed reducing rate at the location of the reference bucket pin P2, and control the speed of the bucket 13c by reflecting the location change amount in the height direction of the bucket pin P2 thereto.

[0103] With reference to FIG. 6, a method for controlling according to the location of the bucket pin P2 during bucket digging will be described as shown below.

[0104] The shortest distance d between the bucket end P1 and the work area is the all the same in postures (a) and (b), but the time it takes for each posture to invade the work area while operating the bucket 13c is different.

[0105] Nevertheless, if the speed of the bucket 13c is limited on the basis of the shortest distance d between the bucket end P1 and the work area, the speed of the bucket 13c is limited all the same in postures (a) and (b). Accordingly, the work speed and efficiency may decrease during the digging work using the bucket 13c.

[0106] Therefore, the speed of the bucket 13c needs to be controlled by reflecting the location change amount in the height direction of the bucket pin P2 even if the shortest distance d between the bucket end P1 and the work area is the same.

[0107] For example, when explaining the case in which the electronic control unit 500 recognizes the location of the bucket pin P2 in posture (a) as a reference location, the electronic control unit 500 sets the speed limit range and/or speed reducing rate on the basis of the shortest distance d between the bucket end P1 and the work area in posture (a).

[0108] In addition, when the posture of the bucket 13c is changed from posture (a) to posture (b), posture (b) refers to a situation when there is a lot of time to invade the work area or when the bucket does not invade the work area as compared with posture (a). In other words, it refers to a case where the distance between the bucket end P1 and the work area is the same, but the location of the bucket pin P2 is raised to be higher than the reference location as much as d_a .

[0109] At this time, the electronic control unit 500 reduces the speed limit amount of the bucket 13c in proportion to the distance d_a raised to be higher than the reference location. In other words, in FIG. 6, the speed reducing graph of the bucket 13c in posture (a) moves in parallel to the speed reducing graph of the bucket 13c in posture (b) as much as the distance d_a raised to be higher than the reference location.

[0110] Accordingly, even if the shortest distance d between the bucket end P1 and the work area is the same in postures (a) and (b), the speed limit amount of the bucket 13c in posture (b) is reduced even more as compared with the reference location, posture (a). In addition, the speed limit range of the bucket 13c in posture (b) is reduced even more as compared with the reference location, posture (a).

[0111] As such, when the electronic control unit 500 sets the location of the reference bucket pin P2, and controls the speed of the bucket 13c by reflecting the location change amount in the height direction of the reference bucket pin P2, it becomes possible to operate the bucket 13c more efficiently than when the bucket 13c does not invade the work area or when there is a lot of time to invade the work area according to the posture of the bucket 13c during bucket digging.

[0112] In addition, the driver may easily operate the work device regardless of driving experience.

[0113] The foregoing description of the present invention has been presented for illustrative purposes, and it is apparent to a person having ordinary skill in the art that the present invention can be easily modified into other detailed forms without changing the technical idea or essential features of the present invention.

[0114] The scope of the present invention is presented by the accompanying claims, and it should be understood that all changes or modifications derived from the definitions and scopes of the claims and their equivalents fall within the scope of the present invention.

DESCRIPTION OF REFERENCE NUMERALS

[0115] **10**: construction equipment [0116] **11**: lower traveling body [0117] **12**: upper swing body [0118] **13**: work device [0119] **13a**: boom [0120] **13b**: arm [0121] **13c**: bucket [0122] **14**: boom cylinder (working actuator) [0123] **15**: arm cylinder (working actuator) [0124] **16**: bucket cylinder (working actuator) [0125] **100**: control valve [0126] **200**: operation lever [0127] **300**: location information provision unit [0128] **310**: location measuring unit [0129] **320**: posture measuring unit [0130] **330**: coordinate calculating unit [0131] **400**: work setting unit [0132] **410**: display [0133] **500**: electronic control unit [0134] **P1**: bucket end [0135] **P2**: bucket pin [0136] **P3**: arm pin [0137] **d**: shortest distance between bucket end and work area [0138] **d s**: distance from bucket end to work area in the direction in which the bucket end moves (the speed direction of the bucket end)

Claims

1. A construction equipment comprising: a lower traveling body; an upper swing body rotatably supported on the lower traveling body; a work device comprising a boom, an arm, and a bucket moved by means of respective hydraulic cylinders and supported by the upper swing body; a control valve for controlling the hydraulic cylinder; an operation lever for outputting an operation signal corresponding to the operation amount of a driver; a work setting unit capable of setting and/or selecting the work area of the work device; a location information provision unit for collecting and/or calculating location information of the work device and/or location information of the work area; and an electronic control unit for outputting a control signal for the control valve according to a signal inputted from at least one from among the control lever, the work setting unit, and the location information provision unit, wherein the electronic control unit is configured to calculate the distance between the work area and the work device and to control the speed of the work device on the basis of the calculated distance, wherein the electronic control unit sets the speed limit range and/or speed reducing rate on the basis of the shortest distance between the bucket end and the work area in a reference bucket pin location.
2. The construction equipment of claim 1, wherein the electronic control unit determines whether the work device approaches or moves away from the work area when an operation signal of the operation lever is input, and limits the speed of the work device only when the work device approaches the work area.
3. The construction equipment of claim 1, wherein the electronic control unit controls the speed of the work device on the basis of the speed reducing rate when the work device enters the speed limit range.
4. The construction equipment of claim 3, wherein the electronic control unit reduces the speed limit amount of the work device in proportion to the raised distance of the bucket pin when the location of the bucket pin is raised to be higher than the location of the reference bucket pin.
5. The construction equipment of claim 1, wherein the location information provision unit comprises at least one of a location measuring unit for measuring the location information of the construction equipment, a posture measuring unit for measuring the posture information of the construction equipment and the location of the respective work device, and a coordinate calculating unit for calculating the coordinate on the basis of the location information measured from the location measuring unit and the posture measuring unit.
6. The construction equipment of claim 1, wherein the operation lever generates an electric signal in proportion to the operation amount of a driver and provides the same to the electronic control unit as an electric joystick.
7. The construction equipment of claim 1, wherein the work setting unit provides a plurality of work mode setting functions that can be set according to the driver's need, and displays, on a display screen, at least one of the geographic information, location information and posture

information of the construction equipment provided from the location information provision unit according to the work mode setting.
